

HARIGOVIND P

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Profile

Recent graduate in Artificial Intelligence and Machine Learning with strong programming fundamentals in Python, Java, and SQL and hands-on experience across the full software development lifecycle. Skilled in building backend systems and RESTful APIs, developing and training machine learning and deep learning models, and designing AI-powered applications including Generative AI and RAG-based systems. Comfortable working with cloud platforms, databases, and version control, with a track record of clean coding, structured testing, and clear technical documentation. Published IEEE researcher with strong analytical thinking and a genuine drive to build reliable, well-engineered software.

Education

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

Bachelor of Technology in Artificial Intelligence and Machine Learning

2022 – 2026

Avadi, Tamil Nadu

CGPA: 8.3

Publications

Real-Time Water Leak and Usage Monitoring for Smart Apartments using IoT Flow Sensors and LSTM-based Leak Detection · IEEE IDCIoT-2026 · DOI: 10.1109/IDCIOT67589.2026.11455854

Technical Skills

Programming Languages: Python, Java, SQL, C

Backend and APIs: REST APIs, FastAPI, Flask, Django

Web Technologies: HTML, CSS, JavaScript

Machine Learning and Deep Learning: PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Supervised Learning, Unsupervised Learning, Classification, Regression, Clustering, NLP

Generative AI and LLMs: Large Language Models, RAG Pipelines, Prompt Engineering, AI Agents, Vector Databases, FAISS

Databases: MySQL, SQLite, MongoDB

Cloud and Infrastructure: AWS, Docker, Linux, Git, GitHub, CI/CD, GitHub Actions

Software Development: Software Development Life Cycle, Agile, Unit Testing, Test Case Design, Debugging, Bug Tracking, Code Review

Projects

Adaptive Knowledge Intelligence System (AKIS)

March 2026

GitHub

- Built an AI system that retrieves answers from a document store using FAISS for semantic search and BM25 for keyword search combined, then verifies every claim in the response against the source text to eliminate hallucinations.
- Exposed the system through a FastAPI REST backend with a Streamlit interface, with structured JSON logging at every stage for full traceability.

Real-Time Violence Detection From Video Using CNN and LSTM

April 2025

GitHub

- Built a model that classifies video clips as violent or non-violent using a CNN to extract visual features per frame and an LSTM to track how those features change across the sequence.
- Trained on a balanced dataset of 1,000 videos, reaching 91% accuracy after iterating on the architecture based on misclassified clips.

GitHub

- Built a Django REST API that automatically matches farmers to the right agricultural expert by converting their problem description into TF-IDF vectors and finding the closest match using k-NN.
- Handled edge cases like sparse input and ranking ties, and ran integration tests to confirm correct matches before deployment.

Certifications

Professional Certificate in SQL and SQL for Data Analysis (Udemy) — Verify

AWS Academy Graduate: AWS Academy Cloud Foundations — Verify

Mastering Generative AI and ChatGPT (GeeksforGeeks, Aug 2025) — Verify

Django Essentials: Build and Deploy Real-World Apps (Udemy, Jul 2025) — Verify